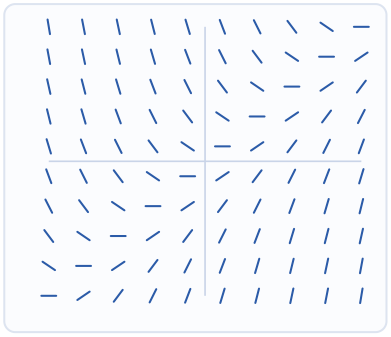
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

First-Order Separable Equations & Existence Theory Practice Exercises

1 MCQ

The second-order linear homogeneous ODE $f''(x) - 39f'(x) + 350f(x) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.



A 39
B 24
C 42
D 53

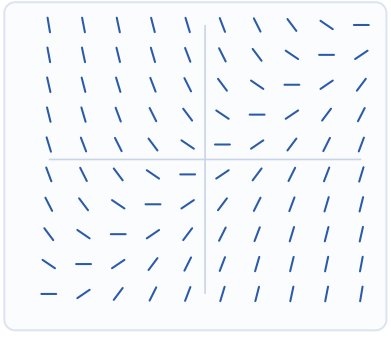
2 MCQ

Consider the linear first-order differential equation $\frac{dy}{dx} + 2y = 6$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

A 2
B $e^2 - 1$
C $3 - 2e^{-2}$
D $3 - 2e^{-1}$

3 MCQ

Consider the differential equation $\frac{dy}{dx} = x + y$ with the initial condition $y(0) = 1$. Use the Euler method with a step size of $h = 0.1$ to approximate $y(0.2)$.



A 1.200
B 1.210
C 1.230
D 1.220

4 MCQ

Consider the second-order linear homogeneous differential equation $y'' - 5y' + 6y = 0$. If the auxiliary equation has roots r_1 and r_2, find the value of $r_1 + r_2$.

A 5

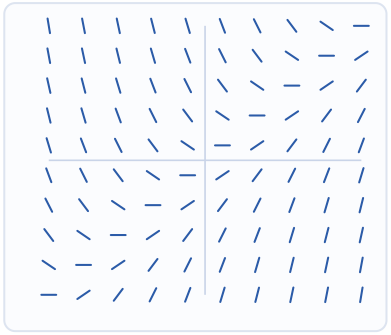
B -5

C 6

D -6

5 MCQ

Consider the separable differential equation $\frac{dy}{dx} = 2x$ with the initial condition $y(0) = 3$. Find the value of $y(2)$.



A 7

B 6

C 8

D 5

6 MCQ

Consider the differential equation $y'' - xy = 0$. The point $x = 0$ is an ordinary point. If we assume a power series solution of the form $y(x) = \sum_{n=0}^{\infty} c_n x^n$, find the recurrence relation for the coefficients c_n.

A $c_{n+2} = \frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

B $c_{n+2} = \frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

C $c_{n+2} = -\frac{c_{n-1}}{(n+2)(n+1)}$ for $n \geq 1$

D $c_{n+2} = -\frac{c_n}{(n+2)(n+1)}$ for $n \geq 0$

7 MCQ

The second-order linear homogeneous ODE $u''(s) - 40u'(s) + 144u(s) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 44

B 43

C 24

D 40

8 MCQ

The Laplace transform of $u(s) = 170e^{19s}$ is $v(s) = \dfrac{170}{s - 19}$ for $s > 19$. Determine the evaluated transform $v(20)$.

A 170

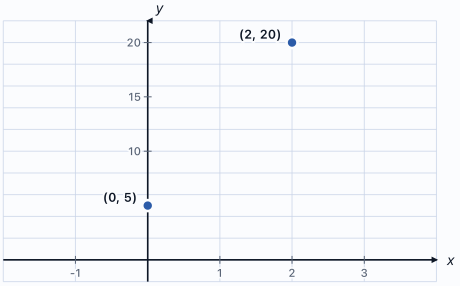
B 151

C 173

D 187

9 MCQ

Consider the separable differential equation $\dfrac{dy}{dx} = ky$ where k is a positive constant. A solution curve passes through the points $(0, 5)$ and $(2, 20)$. Determine the value of k and then find the value of y when $x = 3$.



A $k = \ln 2, y(3) = 40$

B $k = 2\ln 2, y(3) = 40$

C $k = \ln 4, y(3) = 80$

D $k = \ln 2, y(3) = 20\sqrt{2}$

10 MCQ

The second-order linear homogeneous ODE $h''(u) - 42h'(u) + 272h(u) = 0$ has characteristic roots λ_1, λ_2. Determine the trace sum $\lambda_1 + \lambda_2$.

A 45

B 50

C 23

D 42

11 MCQ

Consider the separable differential equation $\dfrac{dy}{dx} = \dfrac{2x}{y}$ with the initial condition $y(1) = 2$. Find the value of $y(3)$.

A $2\sqrt{5}$

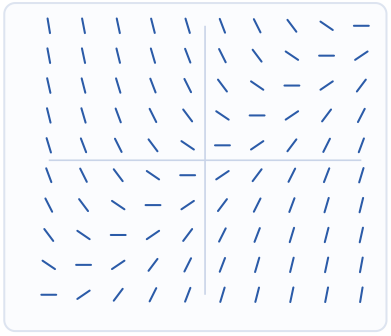
B $2\sqrt{3}$

C $2\sqrt{2}$

D 4

12 MCQ

Consider the differential equation $\frac{dy}{dx} = kx(y^2 - 1)$, where k is a positive constant. If the solution passing through $(1, 2)$ also passes through $(2, 3)$, determine the value of the constant k.



A $\frac{1}{3}\ln(\frac{3}{2})$

B $\frac{1}{4}\ln(\frac{2}{3})$

C $\frac{1}{4}\ln(\frac{3}{2})$

D $\frac{1}{3}\ln(\frac{2}{3})$

13 MCQ

The separable initial value problem $\frac{dh}{du} = 30u$ satisfies $h(0) = 3$. Determine the solution value $h(3)$.

A 141

B 138

C 153

D 135

14 MCQ

The first-order differential equation $\frac{dh}{du} = 23u - 2h$ has its slope field plotted below. Determine the slope $\frac{dh}{du}$ at the point $X(14, 4)$.

A 317

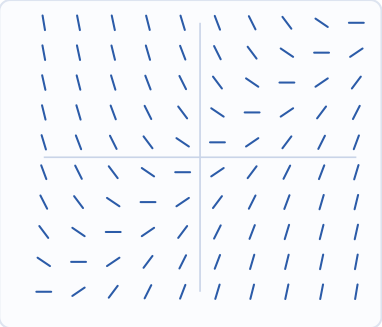
B 310

C 337

D 314

15 MCQ

Consider the separable first-order differential equation $\frac{dy}{dx} = \frac{x^2}{2y}$. If the solution passes through the point $(2, \sqrt{5})$, find the explicit solution for $y(x)$ assuming $y > 0$.



A $y = \sqrt{\frac{x^3}{3} + \frac{7}{3}}$

B $y = \sqrt{\frac{x^3}{3} + 2}$

C $y = \sqrt{\frac{x^3}{3} + \frac{2}{3}}$

D $y = \sqrt{\frac{x^3}{3} + \frac{5}{3}}$

16 MCQ

The separable initial value problem $\frac{dU}{dv} = 28v$ satisfies $U(0) = 9$. Determine the solution value $U(16)$.

A 3584

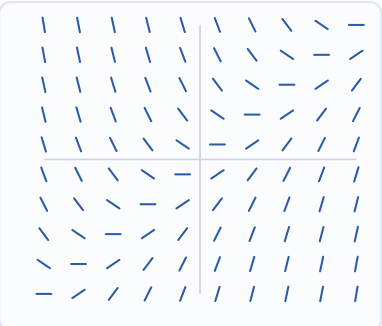
B 3607

C 3596

D 3593

17 MCQ

The separable initial value problem $\frac{dP}{dr} = 34r$ satisfies $P(0) = 8$. Determine the solution value $P(7)$.



A 858

B 844

C 841

D 833

18 MCQ

The first-order differential equation $\frac{dR}{dm} = 9m - 5R$ has its slope field plotted below. Compute the slope $\frac{dR}{dm}$ at the point $M(15, 24)$.

A 18	B -9
C 15	D 24

19 MCQ

The first-order differential equation $\frac{dh}{du} = 21u - 3h$ has its slope field plotted below. Compute the slope $\frac{dh}{du}$ at the point $X(18, 22)$.

A 312	B 290
C 333	D 315

20 MCQ

The first-order differential equation $\frac{du}{ds} = 17s - 7u$ has its slope field plotted below. Compute the slope $\frac{du}{ds}$ at the point $K(2, 24)$.

A -158	B -117
C -134	D -131

21 MCQ

The first-order differential equation $\frac{dF}{dw} = 12w - 4F$ has its slope field plotted below. Compute the slope $\frac{dF}{dw}$ at the point $D(6, 4)$.

A 52	B 59
C 68	D 56

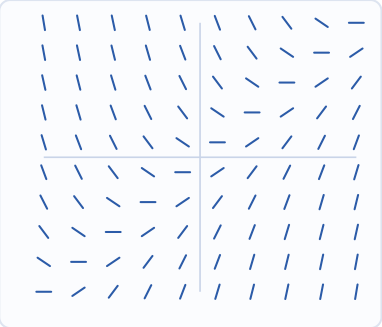
22 MCQ

The first-order differential equation $\frac{dM}{dz} = 11z - 1M$ has its slope field plotted below. Compute the slope $\frac{dM}{dz}$ at the point $G(13, 26)$.

A 120	B 91
C 117	D 128

23 MCQ

The first-order differential equation $\frac{dS}{db} = 4b - 5S$ has its slope field plotted below. Determine the slope $\frac{dS}{db}$ at the point $W(7, 17)$.



A -57	**B** -53
C -54	**D** -74

24 MCQ

The first-order differential equation $\frac{dp}{dt} = 24t - 6p$ has its slope field plotted below. Compute the slope $\frac{dp}{dt}$ at the point $P(17, 25)$.

A 233	**B** 258
C 261	**D** 282

25 MCQ

The first-order differential equation $\frac{df}{dx} = 4x - 2f$ has its slope field plotted below. Compute the slope $\frac{df}{dx}$ at the point $A(10, 21)$.

A -2	**B** 2
C 1	**D** -23